अधिव्याख्याता (इलेक्ट्रॉनिक्स) महाराष्ट्र शिक्षण सेवा, गट-ब
(महाविद्यालयीन शाखा)
उच्च व तंत्र शिक्षण विभाग
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परीक्षेचे टप्पे: १) लेखी परीक्षा - २०० गुण २) मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (११४६)	इंग्रजी	१००	२००	एक तास	पदव्युत्तर पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयाशी संबंधित घटक यामध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ.क्र.	घटक व उपघटक
1	**Semiconductor Devices** Energy bands in solids, concept of effective mass, density of states, Fermi levels Classification: conductors, semiconductors, insulators; intrinsic and extrinsic semiconductors, PN Junction, Forward and reverse biasing of p-n junction, Diode equation and diode equivalent circuit, Breakdown in diodes, Types of Diodes: Zener diode, Tunnel diode Metal semiconductor junction: Ohmic and Schottky contacts Bipolar Junction Transistors (BJT): NPN and PNP transistors, working principle, Transistor Output characteristics and various regions of operation UJT, JFET, MOSFET: Structure, Working Principles, and Output Characteristics High Electron Mobility Transistor (HEMT): Working Principle Optoelectronic devices: LED, Photodiode, Phototransistor, Solar cells: I-V characteristics, fill factor and efficiency LED, LCD and flexible display devices.
2	**Analog Electronics** Rectifiers: Half-wave, full-wave, Bridge and related parameters Voltage regulated Power Supply, Voltage Regulator ICs BJT biasing: Biasing of Bipolar junction, DC and AC Load line, Operating point and stability, BJT as amplifier and as a switch BJT as amplifiers: Configurations: C-B, C-C, C-E, Different Biasing methods of C-E amplifier: Base bias, collector feedback bias and voltage divider, Design of Single stage and Multistage C-E amplifiers. Classification of BJT Amplifiers: Class A, B, AB and C Oscillators: Concept of feedback: positive and negative feedback, Barkhausen criteria Type of oscillators: Wein bridge, Hartley, Colpitt's and Phase Shift oscillators Operational amplifiers (OPAMP): Ideal and Practical parameter values of OPAMP, Block

	diagram of OPAMP, Differential amplifier, Configurations of OPAMP: Inverting and Non-inverting amplifier, Applications of OPAMP: Voltage follower, Adder, Subtractor, Integrator, Differentiator, Comparator, Schmitt Trigger, Instrumentation amplifiers, Wave shaping circuits, Phase locked loops, Active filters, Multivibrators, Voltage to frequency convertors (V/F), frequency to voltage convertors (F/V).
3	**Digital Electronics** Number systems: Decimal, binary, hexadecimal and their interconversions, BCD code, ASCII and EBCDIC codes, 1's & 2's complement subtraction methods Logic Gates: basic, derived, universal gates; Boolean algebra, De Morgan's theorems, minimization techniques (K maps), SOP and POS form of Boolean equations, Combinational Logic Circuits: Half-adder, Full adder, Multiplexers and Demultiplexers, Encoders, Decoders Sequential Circuits: Flip-flops, Memories, Counters: Ring, Ripple, Synchronous, Asynchronous; Shift registers, Programmable Logic Devices (PLD), CPLD, Sequential PLD, FPGA, A/D and D/A converters Digital Circuit Design: Analysis and Design of fundamental mode state machines: State variables, State table and State diagram, Design and Analysis of digital circuits using HDL. Logic Families: Bipolar and unipolar families, TTL and CMOS circuits and its characteristics
4	**Fabrication and Characterization of Devices** IC fabrication: crystal growth, epitaxy, oxidation, lithography, doping, etching, isolation methods, metallization, bonding, Thin film deposition, characterization Techniques: XRD, TEM, SEM, EDX; Thin film active and passive devices, MOS technology and VLSI, scaling of MOS devices, NMOS and CMOS structures and fabrication, Characteristics of MOS transistors and threshold voltage, NMOS and CMOS inverters, Charge-Coupled Device (CCD): structure, charge storage and transfer, Basics of VLSI design, stick diagrams, Layout design rules.
5	**Network Analysis, Signals and systems** Concept of voltage source and current source, impedance of the source and its significance, Network Theorems: Thevenin, Norton, Superposition and Maximum Power Transfer Theorems, Network elements, Network graphs, Nodal and Mesh analysis. Laplace Transform, Fourier Transform and Z-transform. Time and frequency domain response, Passive filters and Two-port Network Parameters: Z, Y, ABCD and h parameters, Transfer functions, Signal representation, State variable method of circuit analysis, AC circuit analysis, Transient analysis, Zero and Poles, Bode Plots. Continuous time signals, Fourier Series and Fourier transform representations, Sampling theorem and applications, Discrete time signal, Discrete Fourier transform (DFT), Fast Fourier transform (FFT), Basic concepts of digital signal processing, digital filters – IIR, FIR.
6	**Microcontrollers and Embedded Systems** Introduction of Microcontrollers: 8051 for embedded systems, Architecture and register set of Microcontroller 8051; Addressing modes, Instruction set of 8051: Data transfer instructions, Arithmetic instructions, Logic instructions, bit level and byte level control transfer instructions; 8051 assembly programming: stack operations, subroutines, interrupts, 8051 programming as timer/counter, 8051 serial communication, 8051 interfacing RS232, LED/LCD display, Keyboard , Stepper motor. Introduction to Embedded Systems: Definition, Components, Block diagram, Working flow, Characteristics, Classification, Modern trends and Applications, Introduction to Embedded C programming
7	**UNIT-VII: Electromagnetism** Electrostatics: vector calculus, Gauss's Law, Laplace and Poisson's equations, Magnetostatics: Biot Savert's law, Ampere's law; Electromagnetic induction, Maxwell's equations and wave equations, Plane wave propagation in free space, dielectrics and conductors, Poynting theorem, Reflection and refraction, polarization, interference, coherence and diffraction, Transmission lines and waveguides: line equations, impedance, reflections and voltage standing wave ratio, rectangular waveguides. Antennas: retarded potential and Hertzian dipole, half wave antenna, antenna patterns, radiation intensity, gain, effective area and Frii's free space receiver power equation. Microwave Sources and Devices: Reflex Klystron, Magnetron, TWT, Gunn diode, IMPATT

	diode, Crystal Detector and PIN diode. Radar: block diagram of Radar, frequencies and power used, Radar range equation.
8	**Communication Electronics** Elements of communication system, types of communication, frequency spectrum Need of modulation, Analog modulation and demodulation: AM, FM , PAM, PWM PPM. Principle of super heterodyne receiver, Random signals, noise, noise temperature and noise figure, Basic concepts of information theory, Error detection and correction, Digital modulation and demodulation: PCM, ASK, FSK, PSK, BPSK, QPSK and QAM; Time and Frequency-Division Multiplexing, Multiple Access techniques, Data Communications: Modems, Codes, Principles of Mobile and Satellite Communication, Optical communication, Optical sources: LED, spontaneous and stimulated emission, semiconductor Lasers; Detectors: PIN photodiodes, Avalanche photodiodes (APD), Optical fibers: attenuation and dispersion characteristics, Bandwidth, Wavelength division multiplexing. Fundamentals of Internet of Things (IoT), IoT protocols used for communication.
9	**Power Electronics and Control System** Power devices and systems: characteristics of SCR, DIAC, TRIAC, power transistors, Protection of thyristors against over voltage and over current, SCR triggering - dv/dt and di/dt, triggering with single pulse and train of pulses, A.C. and D.C. motors - construction and speed control. Switch Mode Power Supply (SMPS), Uninterrupted Power Supply (UPS). Control Systems: Open loop and closed loop control system, Block Diagram reduction techniques, transfer function and signal flow diagram, Stability criterion: Routh-Hurwitz and Nyquist plot, On-off controller, Proportional (P), Proportional-Integral (PI), Proportional-Derivative (PD), PID controllers. Programmable Logic Controller: Block diagram, Basics of Ladder Programming
10	**Electronic Instrumentation** Sensors and transducers: Classification, selection criteria, Measurement of physical parameters: temperature, displacement, velocity, acceleration, force, torque, strain, magnetic field, flow, high pressure, low pressure, humidity, Signal conditioning circuits: Differential amplifier, instrumentation amplifier Sensors for IoT applications. Equipment for measurement of R, L and C, voltage, current, power, energy, frequency/time, phase Digital Multimeters, CRO: Block diagram, working of CRO, front panel controls, Digital Storage Oscilloscope Function generator: block diagram and working, Spectrum Analyzer, Biomedical Instruments: ECG, EEG, Blood Pressure Measurements, MEMS and its applications

दिनांक – ३० एप्रिल, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग